

Named Discrete Distributions, Pt 1

CSU, Chico Math 350

2026-03-25

outline

Introduction

Bernoulli Distribution (Ch. 3.3)

Binomial Distribution (Ch. 3.3.1)

Geometric Distribution (Ch. 3.3.2)

References

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Discrete Distributions

We've covered the foundations of random variables, probability distributions, expectation, and variance. Now, let's start to look at some special random variables that occur frequently and have specific properties such that warrant their own names. We will look at 6 different types of discrete random variables. For each we will learn the following:

- * How to define the random variable
- * How to identify the parameters and write the distributional notation
- * The formula for the PMF, and how to find theoretical probabilities
- * Formulas for the theoretical mean and variance
- * How to calculate all of the above using R commands (both theoretical, and via simulation)

NOTE: R Commands

In R, the common distributions are defined by their **root** name with 3 different prefixes:

* d to compute $P(X == x)$

e.g.: dbinom, dgeom, dhyper, dnbinom

* p to compute $P(X \leq x)$

e.g.: pbinom, pgeom, phyper, pnbinom

* r to randomly draw N samples from the specified distribution

e.g.: rbinom, rgeom, rhyper, rnbinom

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Concept

The simplest type of experiment is one in which there are only two outcomes (success/failure, live/die, true/false, yes/no etc.).

When running simulations in Chapter 2, you wrote your experiment to get down to a single TRUE/FALSE. You were creating a Bernoulli random variable. This simple, yet fundamental random variable serves as the basis for the rest of the distributions in this chapter.

Random variable: Let X be a random variable that denotes the outcome from a Bernoulli trial with probability of success p . Specifically let $X = 1$ denote a success, and $X = 0$ denote a failure.

NOTE:What is considered a success is entirely up to context. If you are interested in mortality rate for a certain disease, then death would be a “success.”

Bernoulli Properties

Distributional Notation

$$X \sim \text{Bernoulli}(p)$$

PMF

$$P(X = x) = p^x(1 - p)^{1-x}; x = 0, 1$$

Mean & Variance

$$E(X) = p$$

$$\text{Var}(X) = p(1 - p)$$

R Commands

No named commands, but can simulate with

`sample(c(0,1), prob=c(1-p,p))`

or through a Binomial random variable with $n = 1$.

Example 1

A beet seed has been planted, and will either germinate or not. The probability of germination is 0.8, and germination is considered a success. Let X be whether or not a beet seed has germinated.

$$X \sim \text{Bernoulli}(0.8)$$

$$P(X = x) = (0.8)^x(0.2)^{1-x}$$

Let $X=1$ mean germination and $X=0$ not germination.

$$P(X = 1) = (0.8)^1(0.2)^{1-1}$$

$$P(X = 1) = 0.8$$

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Concept

If n independent random variables $X_1, \dots, X_n$ all have the same Bernoulli distribution with probability of success p , then their sum is equal to the number of X_i s that equal 1, and the distribution of the sum is known as a Binomial distribution. Examples include:

- * Toss 5 coins and count the number of heads
- * The number of times in a week a person is late for work, when they have a 10% chance of being late each day, independent of other days.

Random variable: Let X be a random variable that denotes the number of “successes” in a series of n independent Bernoulli trials, each with probability of success p .

Binomial Properties

Distributional Notation

$$X \sim \text{Binomial}(n, p)$$

PMF

$$P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x}; x = 0, 1, 2, \dots, n$$

Mean & Variance

$$E(X) = np$$

$$\text{Var}(X) = np(1 - p)$$

R Commands:

`dbinom(x,size=n,prob=p)` to compute $P(X == x)$

`pbinom(x,size=n,prob=p)` to compute $P(X \leq x)$

`rbinom(N,size=n,prob=p)` to randomly draw N samples from a $X \sim \text{Binom}(n, p)$ distribution.

Example 2

Plant 10 beet seeds, and assume that the germination of one seed is independent of the germination of another seed, and all seeds have a germination probability of $p = 0.8$. Let X be the number of seeds that germinated.

Our parameters are:

$$n = 10, p = 0.8.$$

The PMF is:

$$X \sim \text{Binomial}(10, 0.8)$$

$$P(X = x) = \binom{10}{x} (0.8)^x (0.2)^{10-x}; x = 0, 1, \dots, 10$$

Example 2

Theoretical Approach:

$$E(X) = n * p = 10 * 0.8 = 8$$

$$Var(X) = n * p * (1 - p) = 10 * 0.8 * 0.2 = 1.6$$

Simulated Approach:

```
x <- rbinom(10000, 10, 0.8)
mean(x)
## [1] 8.001
var(x)
## [1] 1.597359
```

Example 3

A coin for which the probability of heads is 0.6 is tossed nine times. Find the probability of obtaining 3 heads. Let X be the number of heads that appear coins are tossed.

$$X \sim \text{Binomial}(9, 0.6)$$

$$P(X = x) = \binom{9}{x} (0.6)^x (0.4)^{9-x}; x = 0, 1, \dots, 9$$

Theoretical Approach By Hand:

$$P(X = 3) = \binom{9}{3} (0.6)^3 (0.4)^6 = 0.0743$$

```
choose(9, 3)*(0.6)^(3)*(0.4)^(6)
## [1] 0.07431782
```

Example 3

Theoretical Approach with R functions:

```
dbinom(3, 9, 0.6)
## [1] 0.07431782
```

Simulated Approach:

```
x <- rbinom(10000, 9, .6)
mean(x == 3)
## [1] 0.0732
```


Exercise 1

10 students are selected at random, and each has a probability of 0.10 of being a Math major. Let X be the number of Math majors selected.

1. What is the probability that at least one student is a math major?
2. What is the expected number of math majors in a random sample of 10 students?
3. What is $Var(X)$?

You'll want to be comfortable with both approaches (theoretical and simulated).

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Concept

Given a series of independent Bernoulli trials, we are accustomed to thinking of n and p as fixed, and x is considered the number of successes for a binomial distribution. Suppose that the problem is turned around though, and the question is asked, how many trials will be required in order to achieve the first success? Put this way, the number of trials is the random variable and number of successes is fixed.

- * How many three pointers can Stephen Curry make before he misses?
- * The probability that a random person who smokes will develop a severe lung condition in their lifetime is about 0.3. How many people do you have to check on before you meet someone with a severe lung condition?

Random variable: Let X be the number of *failures* before the first *success* in a Bernoulli process with probability of success p .

Geometric Properties

Distributional Notation

$$X \sim \text{Geom}(p)$$

PMF

$$P(X = x) = (1 - p)^x p; x = 0, 1, 2, \dots$$

Mean & Variance

$$E(X) = \frac{1-p}{p}$$

$$\text{Var}(X) = \frac{1-p}{p^2}$$

R Commands:

`dgeom(x,prob=p)` to compute $P(X == x)$

`pgeom(x,prob=p)` to compute $P(X \leq x)$

`rgeom(N,prob=p)` to randomly draw N samples from a $X \sim \text{Geom}(p)$ distribution.

Example 4

Professional basketball player Steve Nash was a 90% free throw shooter over his career. Let X be the number of free throws before he misses one (note that here, $p = 0.1$, the probability of missing).

$$X \sim \text{Geom}(0.1)$$

$$P(X = x) = (0.9)^x(0.1); x = 0, 1, \dots$$

1. If Steve Nash starts shooting free throws, how many would he expect to make before missing one?

Theoretical Approach By Hand:

$$E(X) = \frac{1-p}{p} = \frac{0.9}{0.1} = 9$$

Simulated Approach:

```
num.made.shots <- rgeom(10000, 0.1)
mean(num.made.shots)
## [1] 8.9175
```

Example 4

2. What is the probability that he could make 20 in a row before he misses?

Theoretical Approach By Hand:

$$P(X = 20) = (0.9)^{20} * (0.1) = 0.0122$$

Theoretical Approach with R functions:

```
dgeom(20, .1)
## [1] 0.01215767
```

Simulated Approach:

```
mean(num.made.shots == 20)
## [1] 0.0118
```

Exercise 2

A machine that produces a special type of transistor has a 2% defective rate. The production is considered a random process where each transistor is independent of the others.

What is the probability that the 10th transistor produced is the first with a defect? Let X be the number of good transistors before the first defective, so that $X \sim \text{Geometric}(0.02)$.

As before, you'll want to be comfortable with both approaches (theoretical and simulated).

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References

references I

Speegle & Clair. Probability, Statistics, and Data: A Fresh Approach Using R, 2021.

Also adapted from notes by Dr. Roualdes, Dr. Donatello, and Dr. Gray.